# THE SUPPORT OF A SHEAF OF RINGS ON THE SMALL ETALE SITE IS CLOSED

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

*Editorial note.* Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

**Lemma 0.1.** *The support of a sheaf of rings on the small étale site of an algebraic space is closed.*

*Proof.* This is true because (according to our conventions) a ring is 0 if and only if $1 = 0$, and hence the support of a sheaf of rings is the support of the unit section. □

*Date*: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.